

SUBHASH RAM TALLAPANENI | CS23B073

4th year Undergraduate in Computer Science

 srt1037.github.io 

subhashram236 

sRT1037 



EDUCATION

Indian Institute of Technology Madras | CGPA: 9.47/10

B.Tech in Computer Science and Engineering

 Jul 2023 – Jul 2027  Chennai, Tamil Nadu

Sri Chaitanya Junior College | Score: 96.7%

 Apr 2021 – May 2023  Vijayawada, India

Sri Chaitanya Junior School | Score: 100%

 May 2021  Kannur, Andhra Pradesh

SCHOLASTIC ACHIEVEMENTS

- Secured an **AIR 121** in **JEE Adv 2023** among 1.8L+ Students
- Secured an **AIR 310** in **JEE Mains 2023** among 11L+ Students
- National **Top 1%** in IOQM, NSEC 2022
- Secured **Rank 16** among 190K+ Students in AP EAMCET 2023
- Secured **Rank 34** among 0.25M+ Students in TS EAMCET 2023.

INTERESTS ON AI

- Tend to use loops and **graphical** loops.
- Built an agentic coding pipeline at **Rubrik**. 
- Enhanced version of the above pipeline. 
- Overview of AI & AI vs **Humanity & Consciousness** of AI. 

PROJECTS

5-Stage Scalar Pipeline CPU Simulator

Self Project

- Simulated instruction execution in a 5-stage scalar pipeline (IF, ID, EX, MEM, WB) using C++. Takes machine code as input and outputs execution statistics including total cycles, hazard counts (data/control), and other performance metrics.

Greedy-Based Huffman File Compression

Self Project

- Built a lossless file compression tool using greedy Huffman coding with variable-length symbol encoding for compact, efficient storage.

Lab Projects

Course Projects

CS6630 – Secure Processor Microarchitecture

Recovered AES keys from power traces using passive side-channel attacks.

CS3205 – Computer Networks

Built and tested an application-layer protocol with TCP, UDP, and TCP variants.

CS3300 – Compiler Design

Built a compiler for a subset of Java with lexing, parsing, and code translation.

CS3500 – Operating Systems

Implemented xv6 kernel features including system calls, paging, traps, and scheduling.

CS3100 – Paradigms of Programming

Used OCaml and Prolog to explore imperative, functional, and logic programming.

CS2610 – Computer Organization and Architecture Lab

Implemented interrupt handling, context switching, and virtual paging in RISC-V assembly.

EXPERIENCE

Software Engineer Intern

Rubrik

- Worked on enabling multiple backup targets for DR of on-prem Control plane – either NAS or Rubrik's own data plane (CDM) – for restoration in case of an outage, involving **Kubernetes** and Rubrik's core internal products.

Software Engineer Intern

Arcana Analytics

- Built an MCP server in Python exposing tools for AI agents, and optimized latency by **dynamically** loading only the relevant tool subset per request instead of relying on stale TTL-based caching.

Undergraduate Researcher

Prof. Mukulika Maity | IIT Madras

- Working with WiFi **CSI** (Nexmon, PicoScenes) for device-free human activity recognition, using CNN, LSTM, and **Transformer**-based models on processed amplitude, phase, Doppler features.

Teaching Assistant

Prof. Kartik Nagar | IIT Madras

- Assisted in teaching **functional** programming in OCaml and **logical** programming in Prolog.

Prof. Gopalakrishnan Srinivasan | IIT Madras

- Evaluated coursework and exams for undergraduate students on core hardware and systems concepts.

SKILLS & SOFTWARES

- C, C++, Python, OCaml, RISC-V Assembly, Prolog.
- Claude Code, Mini-agentic Frameworks, Obsidian Knowledge Graphs, MCPs.
- PyTorch, Pandas, NumPy.
- Matplotlib, Git, GitHub, LaTeX, Verilog, Kubernetes.

RELEVANT COURSE WORK

- Artificial Intelligence, Multi Agents, GPU Programming.
- Probability, Pattern Recognition and Machine Learning, Deep Learning.
- Discrete Mathematics, Basic Graph Theory, Design and Analysis of Algorithms.
- Compiler Design, Operating Systems, Paradigms of Programming, Computer Architecture and Organization.

POSITIONS OF RESPONSIBILITY

Super Coordinator, Ambience Team, Saarang

- Leading visual identity and on-ground decor for Saarang.

Saathi Mentor

- Helping freshmen navigate campus life.

INTERESTS & HOBBIES

- Exploring ML frameworks, learning patterns, and approaches to building ML systems.
- Playing pool, kabaddi, volleyball & watching podcasts.
- Enjoy playing chess against **Stockfish**.